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Perspective

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# The transition from genomics to phenomics in personalized population health

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## Abstract

Modern health care faces several serious challenges, including an ageing population and its inherent burden of chronic diseases, rising costs and marginal quality metrics. By assessing and optimizing the health trajectory of each individual using a data-driven personalized approach that reflects their genetics, behaviour and environment, we can start to address these challenges. This assessment includes longitudinal phenome measures, such as the blood proteome and metabolome, gut microbiome composition and function, and lifestyle and behaviour through wearables and questionnaires. Here, we review ongoing large-scale genomics and longitudinal phenomics efforts and the powerful insights they provide into wellness. We describe our vision for the transformation of the current health care from disease-oriented to data-driven, wellness-oriented and personalized population health.

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## Introduction

Globally, lifespans have steadily risen over the past two decades[1], resulting in an ageing population that presents with many chronic diseases[2]. Diagnostics and treatments for these conditions are driving steeply rising health-care costs. Many drugs are not effective for most of the population[3,4], partially owing to clinical trials that lack diversity, as well as insufficient probing of individual genetic and molecular differences. Furthermore, measuring quality metrics across health systems has been inconsistent, is increasingly burdensome and reveals major health inequities among various populations[5,6]. Creating a less expensive, more effective and equitable health-care system is challenging, in part because of the inherent diversity and complexity of human physiology whose functions are carried out via complex networks of organs and tissues.

One approach to address these challenges is personalized population health, here defined as assessing and optimizing the health trajectory of each individual (Fig. 1). This assessment requires leveraging new technologies to deeply sample biological systems for an integrated view built from diverse data types. The main idea is simple: signals need to be interpreted in the context of systems and functional outputs. This approach has two primary implications: first, elements within a network can be examined to look at their relationships rather than as separate biomarkers; second, more data will yield a more complete perspective of the systems underlying human biology and phenotypes.

There is an important difference between studying disease and understanding overall health, that is, a beneficial homoeostatic state. Studying a disease typically targets specific systems and only profiles disease-relevant networks. Conversely, an understanding of health requires broad measures of network robustness and diversity within the context of their roles in maintaining a healthy homoeostatic state, as represented by the phenome. Although the genome is relatively static for an individual, the phenome changes continuously over time, requiring repeated profiling. The phenome is inherently multidimensional and measured using different types of technologies.

The ability to frequently profile human phenomes is emerging; however, there is a lack of consensus on the best approach. To fully understand health, we must capture the ever-changing dimensions and influences of our environment and behaviour. Molecular and clinical data, including blood-based measurements using various multi-omics approaches, along with assessments of health behaviours, cognitive and emotional health, and environmental factors, can inform strategies to optimize individual and population health and provide a window into health and disease[7].

In this Perspective, we start by briefly discussing what we mean by 'health' and how this may differ slightly from the common vernacular. We then define the various data types that collectively define and influence the human phenome. Next, we outline how these diverse and multidimensional data can be analysed. We then discuss how phenomic measurements are and will be integrated into clinical practice, highlighting existing challenges and prospects. Finally, we arrive at five fundamental principles to guide our understanding of personalized population health (Fig. 2) and outline a framework to validate the impact of phenomics on individual health. We close by proposing what a personalized population health initiative that adheres to these principles might look like.

## Profiling the many dimensions of health

Health is multidimensional (Fig. 3), with many interdependent axes that collectively make up the current health state of an individual. One can be healthy in one dimension yet unhealthy in another. Herein lies the first fundamental principle of personalized population health (Fig. 2): the multifaceted nature of human biology necessitates measurements that span as many dimensions of health as possible over time.

If health is not approached as the intersection of many interrelated systems, then one risks focusing on and optimizing only a single or a limited number of variables, losing sight of the larger and interconnected impact a variable can have on overall health. Today, health-care systems are divided into specializations, and diagnoses are coded into specific medical 'bins'. Treatment is typically focused on addressing specific symptomatology. In fact, treating a specific condition can increase the risk of another. For example, an individual can routinely control their blood sugar levels using a continuous glucose monitor but fail to achieve proper nutritional balance with regard to lipids, micronutrients or other important health factors. Blood sugar levels are primarily responsive to carbohydrates in the diet, and stable blood sugar levels can be achieved with a very low-carbohydrate diet; however, such a diet can consist of high-saturated fats, which increases the risk of heart disease[8]. The current practice of medicine is, thus, plagued by the unintended consequences of siloing, as has been demonstrated by the inability to treat the diverse disease phenotypes of long COVID-19 effectively[9].

The systematic profiling of the genome and phenome of a person will catalyse the movement to a proactive health-care system, one that does not simply focus on disease care but actively works to keep people well. It takes a systems-level understanding of the multidimensionality of health that integrates heart health, gut health, cognitive health, behavioural health, immune system health and other biological systems. There will undoubtedly be unforeseen developments with this new paradigm, but once clinical pipelines are established, comprehensive data can help make more informed decisions, while minimizing avoidable progression to disease states. This shift in the study of health has the potential to move health-care practices from reactive to proactive.

### Assaying the genome

Three approaches are commonly used to investigate the genome. First, single-nucleotide polymorphism (SNP) chips hybridize the DNA of a person to short, predefined DNA fragments that are bound to a surface to detect specific genetic variants, typically those known to be common in a population, and so sample only a small part of the genome. Second, whole-exome sequencing yields protein-coding sequences but eliminates our ability to find changes in parts of the genome that regulate gene expression and to identify some types of structural rearrangements. Lastly, the most thorough investigation is by whole-genome sequencing, which captures the entirety of the coding and non-coding regions of the genome. Advances in sequencing technology have driven down the cost of genome sequencing using short reads to an amazing US $100 per human genome today[10], revolutionizing our understanding of the ways that genetic variation underlies differential disease susceptibility and response to treatments, now making this approach as feasible as the more targeted methods. Furthermore, although more expensive than short-read sequencing, long-read sequencing technologies, such as single-molecule real-time sequencing and nanopore sequencing, now capture structural rearrangements and epigenetic modifications as well[11]. In addition, other techniques such as molecular combing are available for measuring chromosomal telomere length[12], which has been proposed to be indicative of the lifespan of cells. Moreover, the overall proportion of methylation of DNA can be estimated both in blood cells[13] and for different tissues and cell types[14].

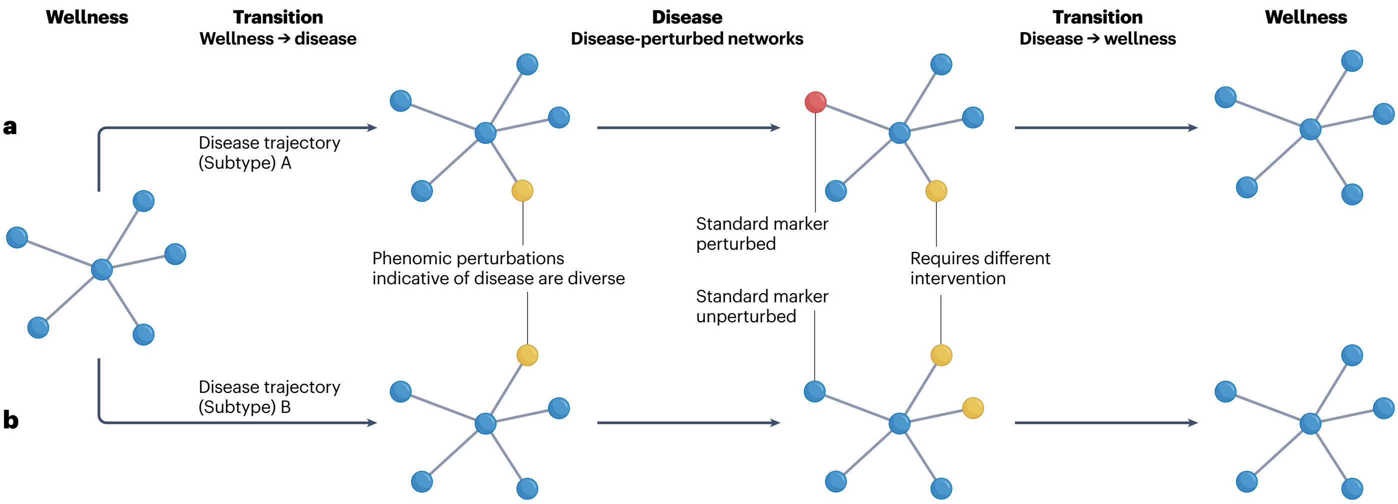



**Fig. 1 | Phenomic snapshots track health trajectories. a**, Phenomic measurements beyond typical clinical measures allow for the early identification of wellness-to-disease transitions. If a disease comprises one or more perturbations to a biological network, then capturing the dynamics of as many networks as possible provides deeper insights into our health status. Throughout our lives, we each undergo many transitions. Optimizing one's health trajectory means identifying – and reversing – wellness-to-disease transitions as soon as possible to prolong health span. **b**, Common disease phenotypes are often the result of different mechanisms (disease 'subtypes'), therefore, probably requiring differing strategies to treat the root cause. The optimal treatment path is different for different disease subtypes, but the subtypes themselves may be undistinguishable by common clinical markers. Expansive phenomic measurements help profile these networks and identify the subtype.

The ability to identify individuals at greater risk of a particular phenotype – using the phenome to contextualize the genome – holds great potential. Stratifying patients into subpopulations based on computed genetic risk factors brings us one step closer to personalized treatment regimens and therapies[11].

## Assaying the phenome

Phenomic data are already used in the clinic, including various clinical laboratory tests (frequently referred to as 'clinical labs') and information about exposures and life experiences. Novel improvements in phenomic data over the past few decades include technology advancements to assay a far greater number of features and the use of systems biology to integrate and interpret these vast data types (Fig. 4). Together, these advances enable deep phenotyping at scale. Technology is also developing to augment multi-omics analysis of individual cells, expanding our depth to single-cell resolution[15]. Here, we provide an overview of the major types of phenomic signals commonly assayed.

**Epigenome.** Although the genome sequence itself is largely static, it is constantly undergoing chemical modifications (for example, the attachment of methyl groups to certain cytosine nucleotides[16]) in various tissues that influence its expression. The technique of sequencing after chemically modifying DNA with bisulfite treatments has long been used to detect these methylated cytosines[17], but some current genome sequencing technologies can detect these and other such modifications directly[18]. Technologies such as chromatin immunoprecipitation followed by sequencing[19] are enabling deeper insights into histone modifications that influence transcriptional control. The epigenome responds to disease perturbations and exogenous environmental exposures[20], allowing for the capture of tissue-specific and time-specific information that has been tied to ageing[21], cancer[22,23] and pregnancy[24], among other areas of human health and development, as well as lifestyle factors such as diet, stress, physical activity and smoking[25]. However, the interpretation of these data requires careful consideration, as the causality link between epigenetic changes and health outcomes can be complex and may vary across individuals and populations[26].

**Proteome.** Only a small number of blood proteins are typically measured in routine clinical laboratories, but plasma contains many thousands of proteins from various organs and tissues. Mass spectrometry continues to be considered by many as the gold standard for precise protein detection and discovery, but more cost-effective and scalable choices for quantification are now available. Recent technological advancements using aptamers[27] or DNA-tagged antibodies[28] used in high-throughput, multiplex immunoassays allow the simultaneous detection of thousands of proteins from serum, plasma and other biofluids or tissue extracts, enabling a systems approach to protein analysis. Although proteomic technologies can identify a wide array of proteins, the sensitivity, specificity and reproducibility of such measurements can vary, especially for low-abundance proteins[7,29]. Interpreting proteomic data requires an understanding of potential confounders, including sample handling, storage conditions and inherent variability among individuals[30].

**Metabolome.** The term 'metabolome' refers to the complete set of small molecules related to cellular metabolism, including lipids, carbohydrates and amino acids. Several metabolites and hormones are measured clinically in blood plasma, urine, saliva or other bodily fluids. This category also includes volatile organic compounds – both endogenous and exogenous – in exhaled breath[31]. Metabolites are chemically diverse and vary in abundance, presenting a challenge for high-throughput metabolomic assays. Although targeted approaches are possible for testing specific hypotheses (including clinical testing) and quantitative precision, a systems-level analysis typically requires an untargeted approach to detect as many metabolites as possible. Mass spectrometry methods form the basis of leading commercial metabolomic applications today[32,33] and allow for the quantification of 1,000–2,000 metabolites, not all of which are present in every individual or every tissue, providing further important information on heterogeneity. Home-based technologies are also on the rise in which, for example, urine biomarkers can be captured and profiled through toilet-installed devices and profiled[34]. Nonetheless, a limitation in distinguishing between truly relevant metabolic changes and benign fluctuations remains, especially given the wide array of factors, from diet to circadian rhythm, that can influence metabolite levels.

**Microbiome.** Pioneering efforts to understand microbial diversity led quickly to an appreciation that the human body hosts an ecosystem of microorganisms that live on and in us, comprising four primary areas: intestinal, skin, vaginal and oral microbiomes[35]. Sequencing amplified marker genes (for example, a gene that encodes 16S rRNA) is common for surveying microbial composition, but metagenomic DNA sequencing allows for investigating the full complement of genomes in a sample. Microbiomics complements metabolomics, with over 200 compounds found in the human blood that are known to be metabolized or exclusively produced by microbiota[36]. The role of the microbiome is emerging as an important factor influencing human physiology, including brain[37], liver[38], immune system[38,39] and digestive[40] health, as well as having a key role in drug efficacy and metabolism[41,42]. Tracking phenomic signals over time helps in enriching existing phenomic data and in understanding trends in the health trajectories of individuals. Being a complex and dynamic ecosystem, microbiome data is limited by large inter-individual variability owing to genetic factors, diet, age and the environment, which makes it challenging to establish a 'normal' or 'healthy' microbiome benchmark[43]. Moreover, the accuracy of microbiome analyses often depends on the quality of reference databases used; many microbial species, especially from non-Western populations, are under-represented or absent in current databases[44].

**Lifestyle factors and physiology.** Digital health spans a broad range of technologies and will help achieve personalized health and telemedicine at scale. Initially, wristwatch-style devices were designed to do little more than count daily steps to encourage exercise, but more recently, new devices, such as cardiac ultrasound devices[45], are providing profound new ways to measure and track large numbers of lifestyle and physiological factors, including heart rate, heart rate variability, sleep patterns and blood oxygenation levels, and can even perform two-point electrocardiograms. The role of 'wearables' in digital health has risen over the past decade. Various devices can measure signals both non-invasively (for example, optically) and from biofluids (for example, sweat or interstitial fluid)[46]. Continuous glucose monitors can be used by individuals with diabetes mellitus and are starting to be

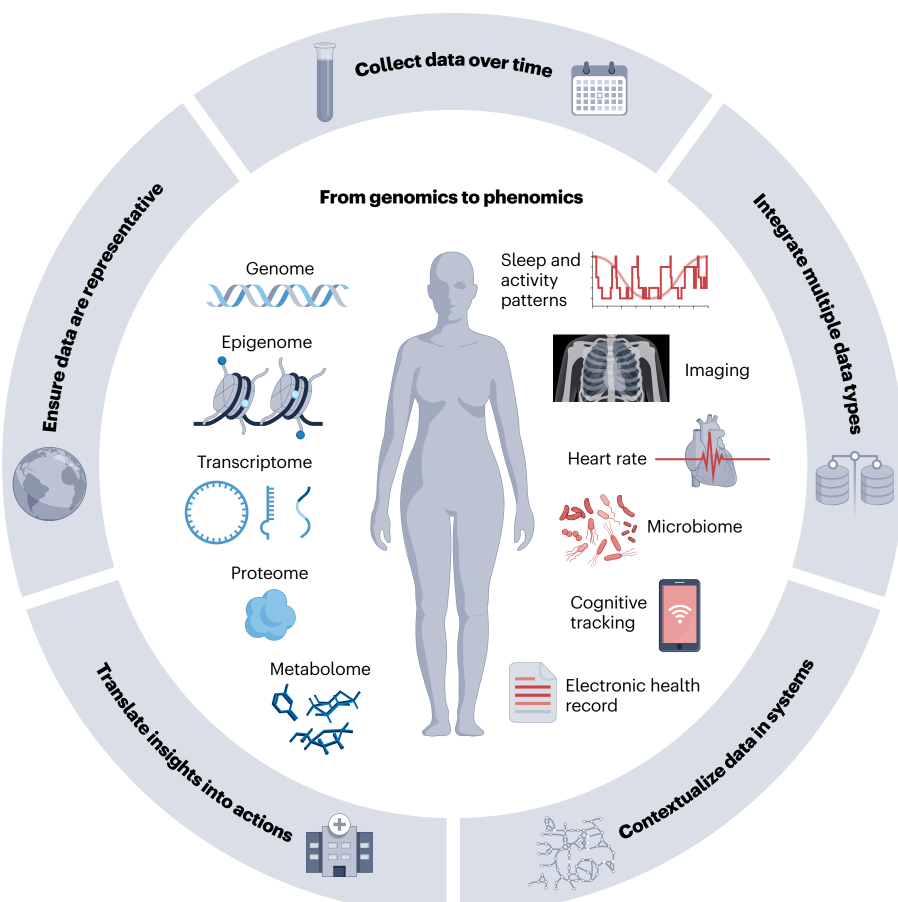



**Fig. 2 | Five fundamental principles of personalized population health.** Emerging efforts around the world are collecting large-scale biomedical data, from genomes to longitudinal phenomes. We propose that these efforts adhere to a set of five principles to guide the design and analysis of these programmes, which ensures the global interoperability and value of both the data and the biomedical findings.

used in some obesity treatment programmes[47]. Furthermore, the number of digital health applications for smartphones has grown over the past decade, a trend that is probably going to continue[48]. Correlating these signals with molecular data allows for the development of candidate digital biomarkers that can provide insights into the health state of an individual, potentially targeting individual organs or systems.

The use of digital health tools is particularly beneficial to assess and monitor brain health, as we have limited ability to directly collect biological samples for omics data generation. Increasingly, heart rate variability is being measured by wearable devices as an indicator of sympathetic and parasympathetic activity. We can gain significant insights through scalable app-based assessments[49] that measure functional domains with known underlying clinical implications[50]. Digital health tools add richness to the data making up an individual's phenome by providing daily physical activity outputs to a person's web of interconnected health variables. However, it is crucial to note that accuracy, longevity and user compliance of wearables can vary, potentially affecting the consistency and reliability of the data collected[51].

**Electronic health records.** Included in the broad definition of digital health is the use of electronic health records (EHRs), which contain collectively captured clinical data based on the interaction(s) of an individual with health-care systems. Blood and urine panels generated from clinical laboratories are often analysed as part of routine clinical care or, most often, diagnostics in response to specific symptom presentation (for example, cardiovascular blood panels). Various types of medical imaging[52] – from cardiac ultrasound[53] to dual-energy X-ray absorptiometry[54] – and corresponding analysis techniques[55] provide a wealth of data on everything from specific organ function to whole-body physiology.

EHRs are severely limited by a lack of data interoperability[56], with the types of data collected generally limited to diagnostic testing, diagnosis codes, procedures and prescribed medications. These data are unequally distributed and lead to data gaps owing to different access to care[57] and the fact that the data of individuals are often siloed across several records[58]. Current challenges include finding better ways to pair data from EHRs – much of which is free text (for example, notes of a physician), imaging, histology, pathology and exposure to toxins – to molecular data to build extensive resources that contain health outcomes[59]. Efforts to preserve the privacy of records while sharing them across organizations (known as privacy-preserving record linkage) are underway[60,61] but will require large-scale efforts and policy changes to become ubiquitous. Equally important are efforts to implement national EHRs in low-income and middle-income countries to address the goals of the World Health Organization and United Nations for ensuring healthy lives and promoting well-being for all people of all ages[62].

## Standards for phenomic data

Upon generating the vast array of multimodal phenomic data described in the previous sections, it is crucial to use standards for the storage and modelling of the data to ensure interoperability and reproducibility.

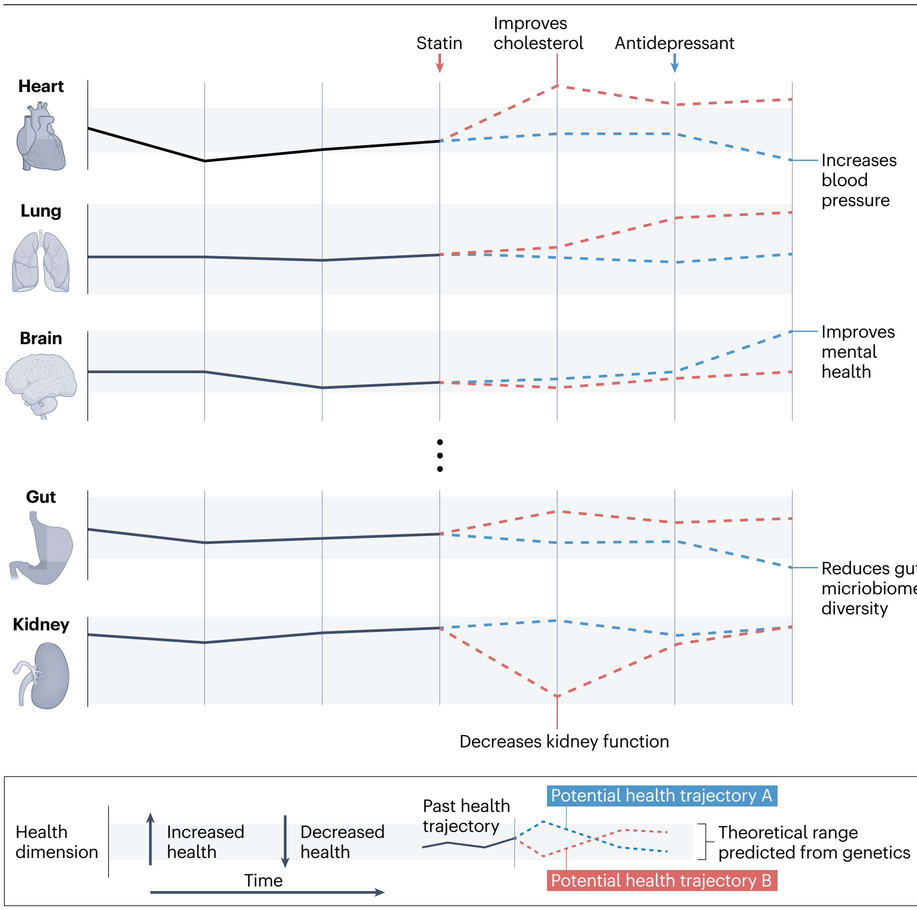



**Fig. 3 | The multidimensional health trajectory of an individual.** Health is a multidimensional continuum across wellness, disease and transitions between them. At any given time, we occupy dimensions of these states. But no one is perfectly well across all domains. Targeting one health domain – for example, by optimizing a single variable – while ignoring consequences in other health dimensions can have a detrimental effect on health. For example, statins can improve cholesterol levels but can have a negative impact on kidney function in some individuals. Similarly, antidepressants can improve mental health at the expense of other health dimensions. Thus, it is important to understand systemic effect(s). Although some dimensions are linked, others can be affected by various stimuli independently. On the basis of previous and current data, the goal is to forecast future trajectories and simulate the response to stimuli.

## Glossary

**Actionable recommendations**
Behavioural or clinical intervention strategies driven by data that can be pursued to reach specific health outcome.

**Aptamers**
Short, single-stranded nucleic acid fragments (DNA or RNA) that selectively bind to specific target molecules such as proteins with variable affinities and off-target binding.

**Deep learning**
A specific machine learning approach inspired by the human brain that uses multi-layered (so-called deep) network architectures.

**Deep phenotyping**
A comprehensive analysis of phenotypic drivers and endpoints, from molecular assays and clinical data to social determinants of health and digital measures of environmental exposures and behaviours.

**Digital biomarkers**
Lifestyle and physiological factors measured via digital health devices that enable remote — and possibly non-invasive — surveillance of the health state.

**Digital health**
Digital tools such as mobile apps, wearables and telehealth that help provide a more holistic view of the health state of an individual.

**Digital twins**
Computable digital, dynamic representations of a human that continuously integrates physiological and biochemical data with mechanistic knowledge.

**Explainable artificial intelligence**
(XAI). Interpretable deep learning that provides explanations for black-box predictions.

**Genetic variants**
Differences in the DNA sequence — from single nucleotides to large regions — among individuals, populations or species that may be inherited (passed down from parents in the germline) or somatic (developed de novo) and can explain differences in physical appearance, disease susceptibility and how people react to pharmacological interventions.

**Health outcomes**
Labels derived from clinical metrics that reflect an individual's dynamic state of physical, mental and social well-being, often used as ground truth (or expected values) to train computational models.

**Health trajectory**
The trend of the integrated health of an individual, which is multidimensional and may simultaneously direct away from one disease and towards another.

**Incidental findings**
Observations that were not the primary objective of the study but could have potential significance or implications for the health of the subjects.

**Knowledge graphs**
Network representations of entities (for example, molecules, diseases and drugs) and the relationships between them.

**Multi-omics**
Multiple omic data modalities, such as genomics, transcriptomics, epigenomics, metabolomics, microbiomics and proteomics.

**Personalized population health**
An approach aimed at optimizing the health trajectory of each individual at the population scale.

**Phenome**
The collection of dynamic and observable characteristics of an organism, ranging from the amount of markers in blood (such as cholesterol levels) to physiological signals (such as heart rate).

**Rare diseases**
One of approximately 7,000 diseases affecting fewer than 1 in 200,000 people (USA) or fewer than 1 in 2,000 people (Europe), of which about 80% are thought to be genetic (so-called Mendelian diseases).

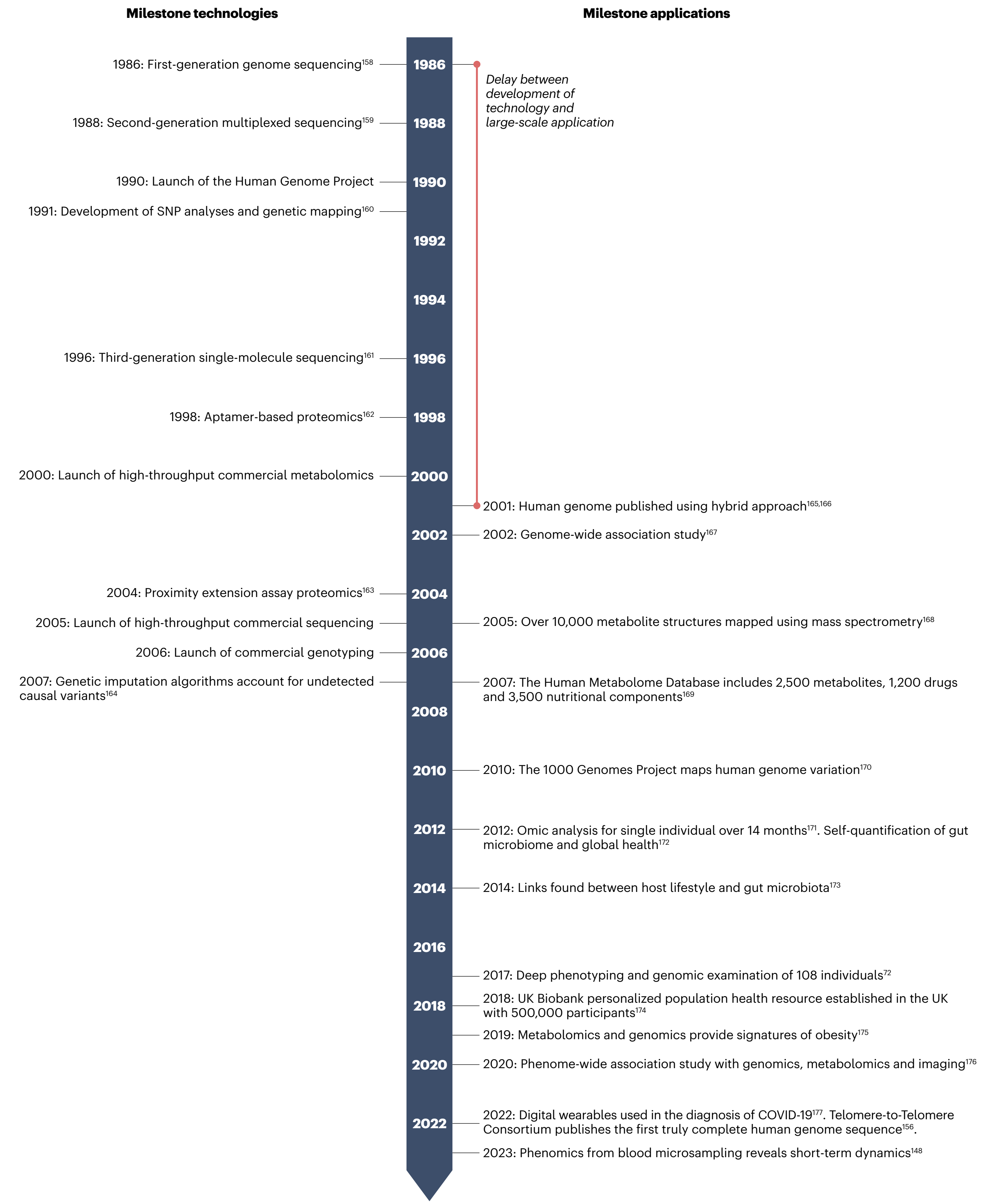
Standards are needed not just for analysis of biospecimens but also for the computational analysis of the data once generated[63]. Extensive community collaboration has resulted in standards for individual data types[64,65] and modelling types[66]. One of the most extensive resources specific to phenomic data is the Human Phenotype Ontology[67], which offers standards for vocabulary, disease phenotype annotations and analysis algorithms. Google's Data Commons has recently been expanded to include biological ontologies via the Biomedical Data Commons[68]. The standards put forth in these types of resources allow for biological entities (for example, genes and proteins) to be linked to one another in mechanistic computational models and knowledge graphs. Moreover, efforts such as the 'Translator' framework of Biomedical Data Translator Consortium perform meta-integration to consolidate various ontologies and bridge terminological gaps[69].

## Integrated analysis of multimodal data

Currently, the various technologies and assays described in the previous sections are mostly research tools and have yet to be optimized for translation to the clinic. Although they empower individuals to measure a great many features, the value of these individually measured quantities and most importantly determining which quantities carry the most actionable information remain a work in progress. The limitations inherent in various phenomic data, particularly related to sensitivity, specificity and sample size, can be addressed and potentially overcome through the integration and triangulation of datasets. This is our second fundamental principle of personalized population health (Fig. 2): the integration of multiple phenomic data types yields vastly more insights than analysing them in isolation. Here, we discuss integrated analysis approaches for large-scale datasets that allow for a holistic understanding of complex systems. Unlike the reductionist approach, these methods enable us to capture a snapshot of a person's biological complex systems and their ability to respond to internal and external signals, such as environmental exposures or diseases.

### Defining metrics to track health

The analysis of multimodal phenomic data is an inherently integrative approach, which yields insight from multiple facets of a complex biological system applied for mechanistic studies of acute or chronic disease[70,71]. For example, in a study of 139 patients with COVID-19 and 258 healthy controls, using data from blood single-cell RNA sequencing, plasma proteomics and metabolomics to guide disease management improved understanding of immunological responses and potential therapeutic intervention points[70,71]. Integrative insights are derived by finding new associations and mechanisms[72] or by discovering reliable and non-invasive disease biomarkers[73,74]. For example, reliable and robust metrics need to be defined that can track the phenome and relate it to commonly used clinical outcomes used by physicians and recorded in EHRs. This ability has been demonstrated using, for example, protein



Milestone technologies
Milestone applications
1986: First-generation genome sequencing[158]
1986
Delay between development of technology and large-scale application
1988: Second-generation multiplexed sequencing[159]
1988
1990: Launch of the Human Genome Project
1990
1991: Development of SNP analyses and genetic mapping[160]
1992
1994
1996: Third-generation single-molecule sequencing[161]
1996
1998: Aptamer-based proteomics[162]
1998
2000: Launch of high-throughput commercial metabolomics
2000
2001: Human genome published using hybrid approach[165,166]
2002
2002: Genome-wide association study[167]
2004: Proximity extension assay proteomics[163]
2004
2005: Launch of high-throughput commercial sequencing
2005: Over 10,000 metabolite structures mapped using mass spectrometry[168]
2006: Launch of commercial genotyping
2006
2007: Genetic imputation algorithms account for undetected causal variants[164]
2007: The Human Metabolome Database includes 2,500 metabolites, 1,200 drugs and 3,500 nutritional components[169]
2008
2010
2010: The 1000 Genomes Project maps human genome variation[170]
2012
2012: Omic analysis for single individual over 14 months[171]. Self-quantification of gut microbiome and global health[172]
2014
2014: Links found between host lifestyle and gut microbiota[173]
2016
2017: Deep phenotyping and genomic examination of 108 individuals[72]
2018
2018: UK Biobank personalized population health resource established in the UK with 500,000 participants[174]
2019: Metabolomics and genomics provide signatures of obesity[175]
2020
2020: Phenome-wide association study with genomics, metabolomics and imaging[176]
2022
2022: Digital wearables used in the diagnosis of COVID-19[177]. Telomere-to-Telomere Consortium publishes the first truly complete human genome sequence[156].
2023: Phenomics from blood microsampling reveals short-term dynamics[148]

**Fig. 4 | Personalized population health enabled by innovative technologies and their large-scale application.** There is a delay between the development of various technologies (top) and their widespread application and adoption (bottom) at a scale necessary for population studies, often because of commercial optimization. First-generation genome sequencing[158]. Second-generation multiplexed sequencing[159]. Launch of the Human Genome Project. Development of SNP analyses and genetic mapping[160]. Third-generation single-molecule sequencing[161]. Aptamer-based proteomics[162]. Launch of high-throughput commercial metabolomics. Proximity extension assay proteomics[163]. Launch of high-throughput commercial sequencing. Launch of commercial genotyping. Genetic imputation algorithms account for undetected causal variants[164]. Human genome published using hybrid approach[165,166]. Genome-wide association study[167]. Over 10,000 metabolite structures mapped using mass spectrometry[168]. The Human Metabolome Database includes 2,500 metabolites, 1,200 drugs and 3,500 nutritional components[169]. The 1000 Genomes Project maps human genome variation[170]. Omics analysis for single individual over 14 months[171]. Self-quantification of gut microbiome and global health[172]. Links found between host lifestyle and gut microbiota[173]. Deep phenotyping and genomic examination of 108 individuals[72]. UK Biobank personalized population health resource established in the UK with 500,000 participants[174]. Metabolomics and genomics provide signatures of obesity[175]. Phenome-wide association study with genomics, metabolomics and imaging[176]. Digital wearables used in the diagnosis of COVID-19 (ref. 177). Telomere-to-Telomere Consortium publishes the truly complete human genome sequence[156]. Phenomics from blood microsampling reveals short-term dynamics[148].

markers identified from deep phenotyping, which indicated transition to various cancers in seemingly healthy individuals months, or even years, before a clinical diagnosis[74].

Newly developed metrics for assessing health include multi-omics-based biological ageing clocks[75] and a multi-omic body mass index (BMI) measure[76]. Biological ageing clocks are machine learning-based tools built from different omics data types that can quantify molecular ageing rates and may help identify modifiable components that can reduce biological age in a personalized manner[77]. For example, increases in biological age relative to chronological age have been associated with disease conditions, but favourable behavioural changes were able to decrease biological age over time[75]. The biological BMI measure — which was based on genomics and longitudinal measurements of metabolomics, proteomics, clinical laboratories, gut microbiomes, wearable devices, and health or lifestyle questionnaires — provided a better prediction of metabolic health than the widely used clinical BMI measure determined from weight and height[76], which misclassifies up to 30% of people[78,79]. People with normal BMI but high biological BMI had worse clinical laboratory measurements, and individuals who lost weight showed variable trajectories in different features of data-driven BMI[76]. The end-goal for these metrics is the development of a robust product that could be implemented in a clinical setting; meeting this goal will require extensive validation on diverse populations, benchmarking against any existing gold standards, sensitivity analyses to determine the robustness of the metrics, and techniques to prioritize based on triangulation from disparate data modalities.

One of the many advantages of multi-omics is our ability to gain novel biological insights into health and disease by integrating large multi-omic datasets. One such example is the integration of two omics platforms: gut microbiome sequencing and blood metabolomics data. The gut microbiome is known to be implicated in multiple health conditions, such as irritable bowel disease[80], gastric cancer[81], *Clostridium difficile* infection[82] and diabetes mellitus[83]. A recent study has shown that metabolites measured in the blood can be used to predict the gut microbiome structure of individuals[73]. Identifying potential microbiome health biomarkers in routine blood tests may open new research possibilities on how the gut microbiome affects human health through blood metabolites. Studies have shown that some microbial metabolites can protect from disease (that is, short-chain fatty acids), whereas others can promote disease (that is, uraemic toxins)[84]. The next step in this field is the concept of engineering the microbiome in a personalized manner, involving the deliberate modulation of the gut microbiota to influence metabolite production according to the specific health requirements of an individual[43]. An important effort in this field is the Nutrition for Precision Health programme, powered by the All of Us Research programme, which strives to predict individual responses to food and dietary patterns. Advances in microbial therapeutics hold the promise of developing a range of non-invasive interventions to personalize wellness promotion and disease prevention[85].

### Phenomic signatures of genomic variation

Genotyping technologies have led to the development of polygenic scores, also known as polygenic risk scores. Polygenic scores can be computed from genotyping data (SNPs) or whole-genome sequencing data and provide insights into various traits (such as height) or complex diseases such as type 2 diabetes mellitus[86] that stem from various factors (that is, polygenic diseases). The application of polygenic scores in the clinic is still early in its evolution[86–88] and represents a critical research gap[89]. Initial studies suggest that the clinical use of polygenic scores could help avoid unnecessary treatments and potential health and financial adverse effects[90]. The greatest value, however, is obtained from the integration of both genomic and phenomic data (Box 1).

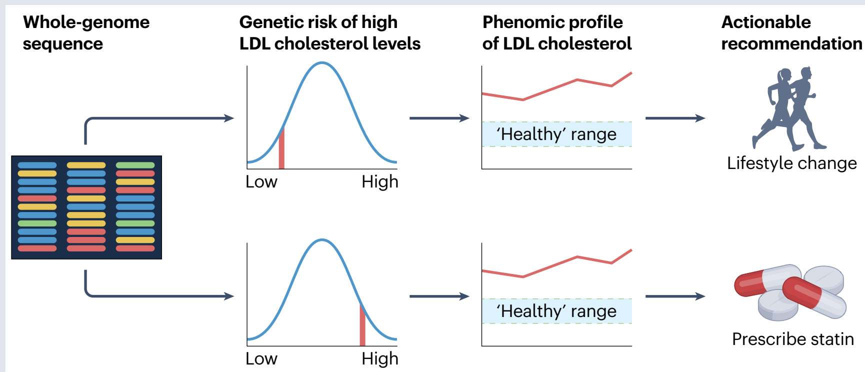
A study of approximately 4,900 individuals has demonstrated that people with elevated risk of various diseases, based on polygenic scores, but without a diagnosis of that particular disease have dozens or hundreds of detectable changes in various phenomic measures in the blood[91]. Many of these observed changes were also present in people with the diagnosed disease, suggesting that these changes, relatively linearly increasing or decreasing with the extent of genetic risk, could potentially serve as markers of early disease progression. The idea that perturbations to blood markers in individuals with elevated genetic risk (computed from polygenic scores) correspond to disease phenotypes is a striking example of how much we can learn about disease from a 'healthy' population. This strategy would suggest that interventions could be developed for high-risk individuals to prevent the manifestation of disease, an important aspect of disease prevention in the future.

One of the biggest areas of potential impact of genomics and phenomics is in the diagnosis of rare diseases, which collectively affect as much as 3.5–5.9% of the world population[92]. A recent study[93] has found that just 379 of the >8,000 rare diseases carry an annual economic burden in the USA of almost US $1 T. Owing to the rarity of each individual disease, many patients undergo a costly 'diagnostic odyssey', with some individuals remaining undiagnosed. A recent study[94] in Europe suggests that early diagnosis of rare diseases could save up to 50% of costs — probably an underestimate in the US health-care system owing to higher testing costs. Approximately 20% of genomic-based diagnoses for rare diseases alter treatment paths[95],

## Box 1

# Translating knowledge to clinical utility through actionable recommendations

All individuals have their own 'normal' range for a given blood marker that is based on their unique genetic makeup. In a system in which the genome of each person is sequenced, all variants are identified and genetic risks of various diseases and health outcomes can be calculated. Because the phenome is a reflection of this genetic backbone, individual markers are contextualized through the genetic risk of a person (see the figure).




Take the example of an individual whose genome is sequenced and examined for the risk of high LDL cholesterol levels. If the individual is found to have a higher genetic risk for elevated levels of LDL cholesterol, then pharmaceutical intervention may be required; but if the individual is found to have a lower risk, then lifestyle modifications may be sufficient to return LDL cholesterol levels to an acceptable range.

As we collect and analyse more data from broader populations, our understanding of risk factors will grow exponentially. Eventually, this will lead to a library of thousands of actionable recommendations driven by data on the genome and phenome of individuals (see the table). Currently, phenome measures such as clinical biomarkers are among the most actionable because they have already been clearly linked to disease trajectories. Adding genetic information to clinical phenomics allows for deeper tailoring of recommendations. The table illustrates several examples of actionable recommendations based on genomic and phenomic data. Tracking overall wellness and the success of various recommendations using various data-driven metrics (such as biological age[75,77,104] or biological body mass index (BMI)[76]) may provide a more holistic view and could supplement the interpretation of individual markers in the future.

| Phenotype | Implicated gene(s) | Scientific rationale | Actionable recommendation |
|---|---|---|---|
| High LDL cholesterol levels in the blood | *LDLR*, *APOB PCSK9* (ref. 178) and polygenic score | High LDL levels are associated with higher risk of heart disease because too much LDL may cause plaques to be formed in your arteries which make arteries less flexible and can lead to clogs | If low genetic risk, prioritize lifestyle interventions: limit saturated and *trans* fats, consume more fibre and increase aerobic activity. If high genetic risk, consult with your physician regarding statins in addition to making healthy lifestyle changes[179] |
| Low DHA levels in the blood | *FADS1* (ref. 180) and *NOS3* (ref. 181) | DHA is a fat that helps control inflammation and is related to heart and brain health | If low genetic risk, increase consumption of foods rich in omega-3 such as cold-water fatty fish (for example, salmon, tuna, herring) to two to three times per week. If high genetic risk, or if dietary changes are insufficient to improve blood levels, add an omega-3 fatty acid supplement containing over 200-mg DHA from fish or algae sources. |
| High ferritin levels in the blood | *HFE*[182] | High ferritin levels are associated with increased risk for diseases such as diabetes mellitus, high blood pressure, heart and liver disease. It is also a biomarker of the genetic iron overload disease, haemochromatosis | Refer to a health-care provider for determination of the cause of elevated ferritin. If the individual is homozygous or heterozygous for the C282Y variant in *HFE*[183], do diagnostic work-up for haemochromatosis |
| Low vitamin D levels in the blood | *GC*[184] | Vitamin D has an important role in bone health, immune function and blood sugar control, and low levels have been associated with heart disease, high blood pressure, diabetes and cancer. Cholesterol-lowering medications (statins), increased age and high BMI can result in lower vitamin D levels | If low genetic risk, recommend increasing consumption of foods rich in vitamin D (fatty fish, liver, mushrooms and egg yolks) or supplemented with vitamin D (orange juice, milk). If high genetic risk, recommend a vitamin D supplement. Some individuals with high genetic risk may need to take up to 5,000 IU vitamin D per day to maintain healthy blood levels |
| High levels of TMAO-producing bacteria in gut microbiome | – | TMAO is an atherogenic metabolite associated with increased risk for cardiovascular events and all-cause mortality[185]. Certain species of gut microbes produce the precursor to TMAO and contribute to elevated blood levels when dietary carnitine and/or choline intake is high | In individuals who consume animal products, gut microbiome results can help reinforce recommendations about the health benefits of reducing intake of red meat and egg yolks, which contain high levels of carnitine and choline. If an individual is vegan, no action is necessary |

(continued from previous page)

| Phenotype | Implicated gene(s) | Scientific rationale | Actionable recommendation |
|---|---|---|---|
| Heartburn | Variants in *CYP2C19* gene affecting response to PPIs[186] | PPIs are commonly taken by patients for heartburn, but some genetic variants alter their effectiveness and adverse effects profile. In some populations, approximately 30% of individuals will have a rapid-metabolizer genotype for PPIs | Recommend alternative, non-PPI approaches to managing heartburn |

DHA, docosahexaenoic acid; PPIs, proton-pump inhibitors; TMAO, trimethylamine *N*-oxide.

significantly improving outcomes in this subset of patients. A recent study has explored rare variant associations with the exome using antibody-based proteomics, demonstrating the potential for proteogenomic data to elucidate possible therapeutic targets[96]. As we generate more genetic data along with multi-omic phenotyping and introduce unexplored dimensions of biomarkers, we have the opportunity to increase the probability of expanding both diagnostic precision and therapeutic options for rare diseases[97], thus furthering the cost savings.

### Next-generation AI and knowledge graphs

The integration of multi-omic data requires advanced computational modelling. The goal is simple: to improve our interpretations and understanding of biological systems. Connecting mechanisms – both known and hypothesized – to data via computational strategies such as artificial intelligence (AI) and knowledge graphs can help us achieve this goal[98]. Understanding the rationale behind the prediction of a model is as important as its accuracy, particularly in human health. As models integrate more complex data, balancing interpretability and accuracy is a challenge. The essence of explainable artificial intelligence (XAI) is to provide not just predictions but also rationales for these predictions. For example, instead of an AI model simply offering a predicted classification of 'cancer positive' or 'cancer negative' for a mammogram, an XAI system would offer extra information explaining why the prediction was made (for example, '1 cm abnormal mass indicates further follow-up'). Because an explanation is given for a prediction, XAI has the potential to aid clinical decision-making by acting as an indicator of abnormal biological processes or of responses to an exposure or intervention[99]. Training AI models is challenging, as health outcomes from noisy data such as EHRs and self-reported questionnaires serve as ground truth[100]. Another challenge is ensuring that knowledge of disease mechanisms and symptoms is included in knowledge graphs to empower XAI approaches[101].

As with any machine learning approaches, XAI performs best in data-rich applications. For example, an explainable framework was developed to reveal the mechanisms underlying adverse drug reactions caused by polypharmacy therapy using a predictive model trained on massive biomedical data (that is, pharmacogenomic knowledge graph and drug–drug interactions)[102]. In a separate study, researchers have developed a multiscale interactome comprising disease-perturbed proteins, drug targets and biological functions[103]. Drug–disease treatments become interpretable via 'diffusion profiles' (that is, biased random walks on the graph based on optimized edge weights) that connect a drug or disease to proteins and biological functions. XAI systems have also been used to calculate and interpret biological age from multiple ageing mechanisms[104]. This system demonstrated good predictive power for mortality, while offering interpretability for the predictions. Similar approaches have shown promise in providing explanations for accurate predictions made by deep learning-based medical image analysis[105,106].

These examples of XAI underscore the importance of incorporating the known relationships between biological analytes into the analytical approach, including the consideration of correlated pathways in the statistical plan (Fig. 5). From a multi-omics perspective, this strategy could lead to several levels of evidence from uniquely generated data sources, strengthening the case for true biological differences without the need for increased sample sizes[107]. For example, suppose five proteins nominally change in expression level between treatment and control: these findings have a higher tendency to be true if they are aggregated in one biological system or pathway rather than dispersed in different systems; they are interrelated variables and, hence, should be analysed using methods that can account for this non-independence (for example, grouping or dimensionality reduction prior to multiple hypothesis correction). Extending this reasoning to thousands of data points makes the compelling argument that more data provide deeper insights into biochemical and physiological mechanisms.

Many next-generation technologies focus on the construction of digital twins[108] for individuals. A digital twin serves as a model of disease or wellness by integrating relevant top–down physiological and mechanistic data with bottom–up molecular, clinical and omics data. One of the important aspects of a digital twin is that it continuously integrates newly collected data, leading to increased predictability of outcomes. A digital twin can create many virtual versions of an individual to test hypotheses. In other words, a specific perturbation can be introduced (for example, a genetic mutation) and the response to many hundreds or thousands of therapies could be tested. A digital twin takes the data from an individual to identify gaps and aberrations and offers suggestions on how to rectify these deficiencies. This is our third fundamental principle of personalized population health (Fig. 2): systems-level analyses will improve data-driven decision-making that relies on analytics and predictive modelling to inform interventions. This approach has been suggested for various areas, such as oncology[108] and cardiovascular disease[109]. Although an ethical discussion is out of scope for this Perspective, the use of digital twins in health-care is not limited to purely scientific impact as their use can have socio-ethical implications[110].

### Applications of phenomic data to commercial or clinical products

Several examples of products that can act as metrics for health and wellness to gather important data exist, including wearables used to diagnose and monitor vitals during infectious diseases[110,111], microbiome tests to monitor gut health and make dietary recommendations[112–114], blood metabolite panels that can robustly predict the diversity of the gut microbiome[73], at-home blood measurement devices connected to

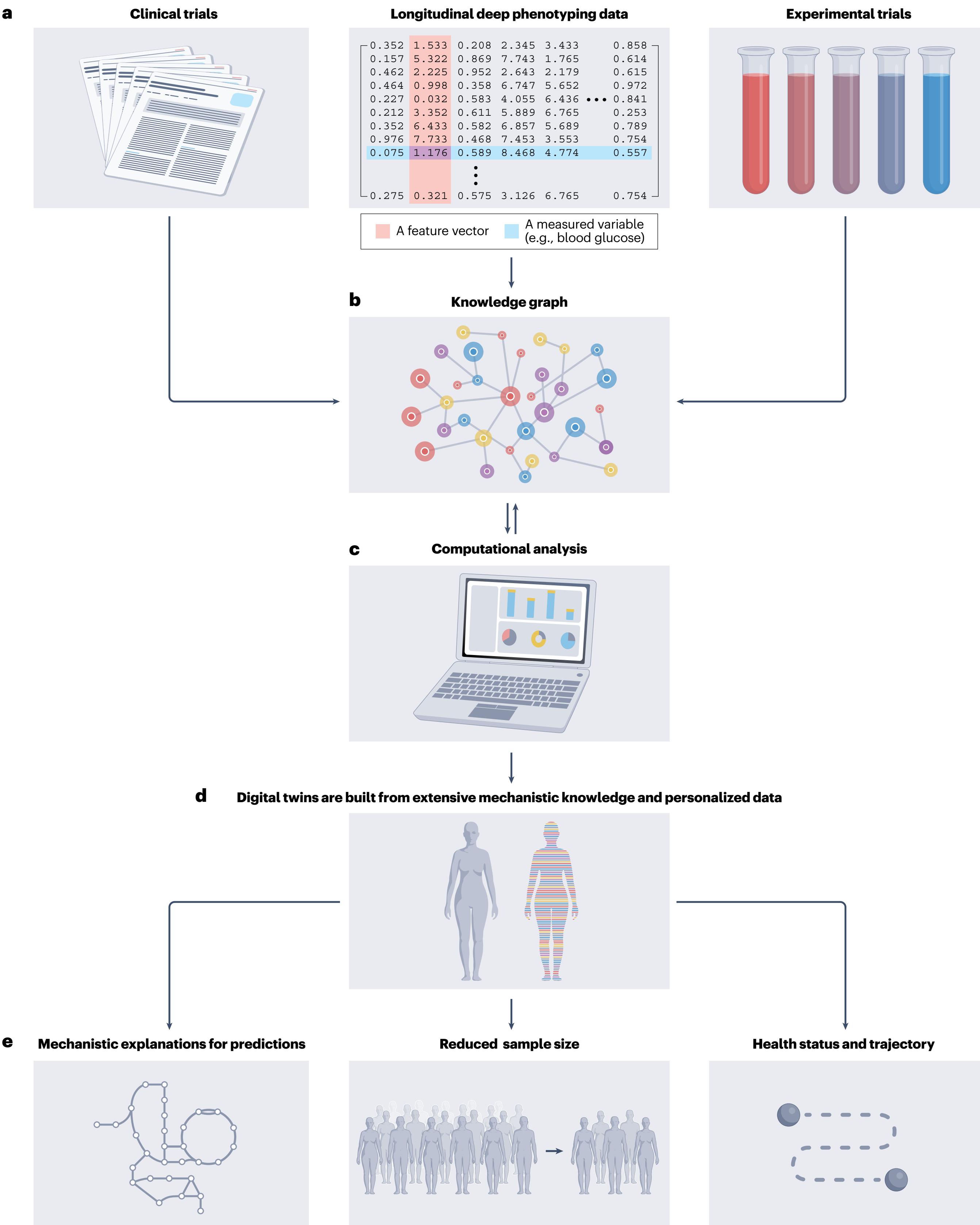

a
Clinical trials
Longitudinal deep phenotyping data
Experimental trials
0.352 1.533 0.208 2.345 3.433 0.858
0.157 5.322 0.869 7.743 1.765 0.614
0.462 2.225 0.952 2.643 2.179 0.615
0.464 0.998 0.358 6.747 5.652 0.972
0.227 0.032 0.583 4.055 6.436 ••• 0.841
0.212 3.352 0.611 5.889 6.765 0.253
0.352 6.433 0.582 6.857 5.689 0.789
0.976 7.733 0.468 7.453 3.553 0.754
0.075 1.176 0.589 8.468 4.774 0.557
0.275 0.321 0.575 3.126 6.765 0.754
A feature vector
A measured variable (e.g., blood glucose)
b
Knowledge graph
c
Computational analysis
d
Digital twins are built from extensive mechanistic knowledge and personalized data
e
Mechanistic explanations for predictions
Reduced sample size
Health status and trajectory

**Fig. 5 | The integration of big data and knowledge graphs enables advanced analytics that surpass correlations and facilitate the identification of causal mechanisms. a**, New analytical techniques allow for the integration of multimodal data types that go beyond simple correlations and yield deep phenotyping data collected over time. **b**, Knowledge graphs are built from computational analyses, experimental data and clinical trial data. **c**, Knowledge graphs are then analysed through classic inferential statistical methods, including correlations and regression models that, in turn, feedback into the knowledge contained within the graph (part **b**). **d**, When deep phenotyping data are integrated with knowledge graphs and analysed, we are able to build digital twins – personalized frameworks for computation. **e**, This framework can help identify causal mechanisms by representing the relationships between different variables in the data. This allows for direct mechanistic explanations behind predictions, reducing the required number of samples and participants, and the tracking and predicting of individual health trajectories (see Fig. 1).

multi-omics platforms[115,116] and various algorithms for biological age[104]. Although a growing number of these exist as consumer products, most are research projects and relatively few have made the transition into the clinic.

## Adopting personalized population health

How can we accelerate the translation of validated technologies into use? This challenge is embodied in our fourth fundamental principle of personalized population health (Fig. 2): complex ideas must be distilled into concise, actionable recommendations that meet evidence-based standards for adoption by clinicians and consumers (Box 1). Digital twins provide an avenue to generate actionable recommendations. Various therapies may be simulated in a virtual environment to gauge the response of an individual, enabling large amounts of data to be contextualized to highly personalized recommendations. As these technologies mature, they will power digital twins capable of offering personalized actionable recommendations.

Recently, large language models (LLMs) and their underlying transformer technology have demonstrated potential in helping deliver personalized population health[117,118]. For example, much useful information and data are contained in EHRs, which are too extensive and not stored in a format conducive to human consumption; LLMs have the potential to facilitate this process for physicians by synthesizing relevant information from the EHR of an individual based on physician queries to help answer questions and identify trends. Such methods are well-suited for effectively articulating highly personalized health recommendations at scale. Unfortunately, there are still great obstacles to overcome with respect to the accuracy and verifiability of such methods[119], including regulatory oversight[120], but there is great promise in their ability to aid medical practitioners in providing personalized care at scale[121].

The two areas that have seen the biggest impact of personalized medicine to date are genomic medicine[122,123] and oncology[107,124]. In 2013, the American College of Medical Genetics (ACMG) published recommendations on the responsible management of incidental findings for patients who have their genomes sequenced[125]: this list has been updated[126] and now includes 82 genes. Mutations in these genes are tied to diseases in which there are clinical actions for a specific mutation, such as in breast cancer and malignant hyperthermia, a pharmacogenetic disorder that manifests as a severe reaction to certain anaesthetics. Beyond the recognized ACMG genes, there are six general classes of actionable recommendations based on genomics. First, pharmacogenomics – linking multi-omic data to drug responses – offers great promise in improving patient outcomes and reducing adverse effects[127]. Second, genomic data can be used to diagnose one of thousands of rare diseases, as discussed earlier. Third, connecting immediate families to extended families via ancestry mapping provides more context for heredity and an understanding of the genetics of disease[128]. Relatedly, a fourth class of recommendations is based on the identification of Mendelian recessive genes[129]. Interestingly, heterozygosity for Mendelian recessives (that is, individuals who harbour a recessive allele) can result in a dampened disease phenotype or even be a risk factor for a different disease[130], and phenomic data beyond the genome can help decipher observable phenotypes[72]. A fifth class is related to lifestyle factors (nutrition, stress, exercise and potential athletic injuries, and sleep), wherein multiple studies have connected SNPs to function and risk[131,132]. Finally, the sixth class comprises polygenic scores showing that high-risk individuals should be treated differently than those at low risk (Box 1).

New personalized cancer therapies have shown promise[107], such as molecularly matched targeted therapy in glioma patients[133]. Many of these therapies are based on pharmacogenomic information that empowers more personalized therapeutics. However, it is important to note that the use of multi-omics and other phenomic data is still limited and in the research phase – the technologies and their various applications are being established. As a result, studies reporting the gold standard against which to benchmark new phenomic approaches have yet to be performed.

### Overcoming barriers to widespread adoption

As we look ahead to the continued implementation of phenomic technologies, one of the greatest challenges will lie in making these technologies accessible to everyone. We lack data from diverse populations. As we have seen from the burgeoning use of genetic sequencing technologies, the value of the insights gained is limited by the input data[134]. This brings us to our fifth and final fundamental principle of personalized population health (Fig. 2): input data need to be representative of the entire population. This is essential to avoid downstream algorithmic bias because of genomic diversity or socioeconomics and to ensure that the benefits of personalization can be made available with high efficacy to all people. For example, studies have shown that the *APOE*-related risk of Alzheimer's disease may not be consistent across different ethnicities[135–137], that there is a higher prevalence of diabetes mellitus in South Asian populations[138,139], and that genetic variation explaining cardiovascular disease[140,141] differs across ethnic groups[142].

Although some of the technologies discussed here have yet to be clinically validated, the primary barrier to their integration into our health-care systems is not scientific – it is economic. The approach we advocate here includes genome sequencing (a relatively mature technology) and longitudinal phenomic profiling (spanning mature to cutting-edge technologies). Although sequencing technology has been broadly accessible for well over a decade, it is still not an integrated part of our health-care systems. Although sequencing costs have steadily dropped, the processes for identification and interpretation of variants and quality checks that are required for clinical use (and that research-grade consumer products do not meet) increase the cost significantly[143,144]. But even for clinically validated genetic applications, clinical professionals lack training in their application, leading to technical barriers with result interpretation[145]. Implementing widespread data-driven technologies, thus, places the onus on researchers to present the output of these new technologies in concise and interpretable ways. A growing number of large-scale personalized population health initiatives are underway around the world (Table 1), but these efforts are primarily research projects and are often not integrated with health-care systems. There is still a large gap between these research initiatives and the translation of their value into clinical practice. Both patients and physicians need to be given the tools to understand the results. Although patients now are generally aware of terms such as 'genes' and that certain mutations in the genome are linked to diseases such as cancers, we must avoid technical jargon and synthesize clear and interpretable results. Phenomic data will not be helpful to patients without proper guidance on how they can make adjustments to improve their own wellness.

Over-testing and over-diagnosing are the common arguments against big data approaches in health science, wherein treating seemingly healthy individuals may require a more careful examination of the cost and adverse effects of interventions. This risk can be mitigated by recognizing that processing vast, multimodal data is fundamentally different from analysing individual or small groups of signals. By analysing longitudinal data from a large research cohort followed over time, we can differentiate between normal variations and early pathological changes. Measuring a multitude of features across a diverse cohort allows for better characterization of the normal variation within subpopulations, further reducing the risk of false positives. Data collected from a person must be contextualized within the networks that make up the physiology of that person, and redundant information can be clustered. One strategy to achieve this is to map the data to mechanistic knowledge graphs or knowledge bases that describe how the systems work and interact with each other[36,146,147]. Moreover, although omics data often do not provide a clear causal link between molecular changes and disease outcomes, collecting longitudinal phenomics is crucial for identifying causality to prioritize potential interventions. One of the big open questions in this area is as follows: when should various phenomic quantities be measured, considering known (for example, circadian rhythms) and unknown dynamics of the underlying signals? New research is providing insights into these questions[148], which will be crucial to the interoperability of existing large-scale phenomic datasets that do not have the same sampling frequencies. Advances in wearable health technology and telemedicine that track sleep, activity, heart rate variability and other health markers can fill gaps in data collection, as they relate to aspects such as

**Table 1 | Examples of personalized population health initiatives currently underway around the world**

| Initiative | Location | Mission | Target cohort size | Scope of data | URL |
|---|---|---|---|---|---|
| All of Us | USA | To build a diverse health database that will help us understand how our biology, lifestyle and environment affect health | 1,000,000+ | Genome (WGS), clinical | https://allofus.nih.gov/ |
| The Emirati Genome Programme | UAE | To explore the genetic makeup of Emiratis, using cutting-edge DNA-sequencing and artificial intelligence technologies that will lead to a personalized and preventive health care | 9,000,000+ (UAE population) | Genome (WGS), clinical | https://u.ae/en/information-and-services/health-and-fitness/research-in-the-field-of-health/the-emirati-genome-programme |
| FinnGen | Finland | To collect and analyse genome and health data from 500,000 Finnish biobank participants | 500,000 | Genome (WGS), clinical | https://www.finngen.fi/en |
| International Human Phenome Project | People's Republic of China | To develop new phenomic technologies and use the resulting data to investigate mechanisms underlying the human phenome | 500,000 | Genome (WGS), behaviour and lifestyle, deep longitudinal phenomics, imaging | https://hupi.fudan.edu.cn/en/index.jsp |
| Million Veterans Program | USA | To study how genes, lifestyles, military experiences and exposures affect health and wellness | 1,000,000 | Genome (WGS and genotyping), behaviour and lifestyle, clinical | https://www.mvp.va.gov/pwa/ |
| oriGen | Mexico | To decipher the Mexican genome and understand precisely the genetic components associated with the most common diseases in the country | 100,000 | Genome (WGS), behaviour and lifestyle, clinical | https://content.tecsalud.mx/proyectoorigen |
| Our Future Health | UK | To build a database linking the health records and genetic information of 5 million people | 5,000,000+ | Genome (WGS), behaviour and lifestyle, clinical | https://ourfuturehealth.org.uk |
| PRECISE | Singapore | To implement data-driven health care, improved patient outcomes and economic value capture | 100,000 | Genome (WGS), behaviour and lifestyle, clinical | https://www.npm.sg/ |
| Project 10K | Israel | To gather information on lifestyle and diseases in the Israeli population | 10,000 | Genome (WGS), behaviour and lifestyle, deep longitudinal phenomics | https://www.project10k.org.il/en |
| UK Biobank | UK | To build a large-scale biomedical database and research resource, containing in-depth genetic and health information | 500,000 | Genome (WGS and genotyping), behaviour and lifestyle, clinical, limited single-time point phenomics | https://www.ukbiobank.ac.uk/ |

UAE, United Arab Emirates; UK, United Kingdom; USA, United States of America; WGS, whole-genome sequencing.

circadian rhythms. Finally, as multi-omics assays become a standard in wellness checks, data collection will no longer be limited to clinic visits.

Opponents further argue that big data has limited discovery potential, especially in the context of omics measurements, which produce numerous features per measurement that often exceed the number of samples. Indeed, low sample-to-feature ratio can result in inadequate power in conventional statistical methods or overfitting of machine learning models. To address these challenges, various methods can be used, such as reducing the number of features through feature selection[149], transforming the original features into a smaller set of relevant information through feature extraction[150], adding a penalty term to the model to prevent overfitting through regularization[73], creating synthetic data by oversampling[151] and using model-based approaches such as Bayesian methods that leverage prior knowledge about the data[152]. Contrary to the notion that more data may lead to less knowledge, appropriate analytical techniques can transform knowledge into power when analysing high-dimensional data.

A major argument against the widespread implementation of personalized population health is the cost of these technologies. Although some commercial programmes exist, they are essentially health and wellness for the wealthy. A whole-genome sequence and twice-annual visits to profile blood proteins and metabolites, gut microbiota, a digital wearable and an extended clinical laboratory panels currently cost upward of US $6,000. Those costs currently fall on individuals because payer systems across the world are not structured to keep people well — instead, they are structured to treat people after the diagnosis of a symptomatic disease. This remains a barrier to accessing and unlocking the power of phenome-centric health care. That said, evidence-based preventive measures that have been implemented have had profound impacts in various disease areas such as the use of vaccines for preventing cervical cancer[153] and statins for preventing cardiovascular disease[154]. The challenge for adoption of additional measurements is to provide compelling scientific evidence and economic data that the benefits will save health-care systems money. Furthermore, we are confident that increasingly broad implementation will provide the market incentives for reducing costs. In the coming years, we envisage increasingly powerful computational techniques that will allow us to greatly reduce the dimensionality of data originally generated for discovery to more tractable numbers of measurements. Indeed, we anticipate an enormous increase in the efficiency and robustness of digital measurements wherein they may act as suitable proxies for biospecimen measurements; we may reach a point in which profiling and assessments will be performed at home, data will be sent to centres for analysis and derivation of actionable recommendations and results will then communicated to clinicians via XAI systems for return of results to individuals.

The data-driven approach to personalized population health that we have discussed here will lead to a new practice of medicine that will transition from its current focus on disease to one of wellness and prevention. To generate the data necessary to address the above concerns with this approach, we and others have proposed personalized population health initiatives based on the fundamental principles outlined here (Table 1); for example, the Human Phenome Initiative will carry out detailed longitudinal phenomic analyses on one million individuals over 10 years. If completed, such an initiative would demonstrate the capacity for phenomic data to drive an increase in the quality of health care while leading to significant cost savings for health-care systems. Furthermore, the economic impact would mirror that of the Human Genome Project[155], significantly bringing down the cost of phenomic technologies by orders of magnitude to a point in which widespread adoption is feasible.

## Conclusions

Just as our understanding of the human genome has continuously improved over the past 20 years[156], so too will our ability to accurately profile and understand the human phenome and how it relates to the genome. Here, we have advocated for the application of phenomic technologies to help make personalized population health a reality. As these technologies improve, we must work to translate them to clinical utility. Part of the continued development of these technologies will include an optimization of sample processing; results need to be returned to participants and patients within hours or days, not months. This influx of data combined with powerful AI approaches (including knowledge graphs, digital twins and LLMs) will lead to actionable recommendations and interpretable results that will empower clinicians to practise medicine like never before, to focus on keeping healthy people well, to prevent disease before it ever manifests and to reverse and cure diseases that do develop.

Existing personalized population health programmes (Table 1) vary in both size and depth of measurements, spanning only genetics to extensive longitudinal phenomic measurements[157]. However, despite the promise of these initiatives, they are still mostly research initiatives. Their impact on our understanding of human health will be extraordinary, but we must take the next step to translate these findings into clinical practice. We also need to further ensure that these programmes are representative of the entire human population and make their data accessible to researchers throughout the world. We hope that the five fundamental principles of personalized population health we have laid out here will help in the continued design, implementation and clinical translation of such programmes:

(1) Taking a systems approach that encompasses as many aspects of the genome and phenome over time as possible is crucial to understanding the complexities of human health.
(2) Integrating multiple phenomic profiling technologies generates significantly greater value than when individual technologies are used in isolation.
(3) Incorporating genomics and phenomics improves data-driven decision-making by building powerful predictive models that inform interventions.
(4) Deriving actionable recommendations from integrating multimodal data is necessary for translating phenomic data to the clinic.
(5) Measuring data that is representative of the diversity within the overall population is vital to contextualize the biological significance of genomic and phenomic variation.

We have a need for a personalized population health initiative built on these foundational principles. We envision tracking a large cohort of 1 million people — one that is representative of the larger human population — for 10 years or more, with frequent deep phenotyping that uses as many of the technologies outlined above as possible. Such an effort would require academic and industry partners working in concert and would result in a truly ground-breaking resource: a database of genomes and longitudinal phenomes that would be accessible to the world for biomedical discovery.

As this field develops, we look to our colleagues to refine these principles and implement these approaches in health-care systems around the globe. Ultimately, understanding the human phenome is vital to improving human health, and collecting large amounts of phenomic data on large and diverse populations is key to advancing personalized population health.

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## Author contributions
J.T.Y., S.J.E., N.R. and J.L.B. researched the literature. J.T.Y., S.J.E., N.R., J.L.B., J.C.L. & L.E.H. contributed substantially to the discussions of the content. J.T.Y., S.J.E., N.R., J.L.B., J.C.L. & L.E.H. wrote the article. All authors reviewed and/or edited the manuscript before submission.

## Competing interests
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